

# CHEYANNE SHARIAT

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## EDUCATION

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**Ph.D., Astrophysics**, California Institute of Technology 2024 -  
**M.S., Astrophysics**, California Institute of Technology 2024 - 2026  
DOE Computational Science Graduate Fellow  
Advisor: Kareem El-Badry

**B.S., Physics**, University of California, Los Angeles 2021 - 2024  
*summa cum laude*  
Tisch Academic Excellence Scholar  
Advisor: Smadar Naoz

## RESEARCH POSITIONS

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**Graduate Student Researcher** 2024 -  
California Institute of Technology *Pasadena, CA*

**Undergraduate Student Researcher** 2021 - 2024  
University of California, Los Angeles *Los Angeles, CA*

**Summer Undergraduate Research Fellow** 2023  
NASA Jet Propulsion Laboratory (JPL) *Pasadena, CA*

**Summer Undergraduate Research Fellow** 2022  
Caltech High Energy Astrophysics Group *Pasadena, CA*

**High School Researcher** 2019 - 2022  
UCLA Neurology and Neurogenetics, Fogel Lab *Los Angeles, CA*

## HONORS & AWARDS

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DOE Computational Science Graduate Fellowship 2025 – 2029

NSF Graduate Research Fellowship (offer declined) 2025

Raynor L. Duncombe Student Research Prize, AAS DDA 2025

Joshua and Beth Friedman Fellowship, Caltech 2025

Dean's Prize for Excellence in Research, UCLA 2024

Lindau Nobel Laureate Meetings Fellow, Lindau, Germany 2024

Helen Quinn Award for Undergraduate Research Theory (1st), APS 2023

Summer Undergraduate Research Fellowship, NASA Jet Propulsion Laboratory 2023

Undergraduate Research Fellowship, UCLA 2022 – 2023

Summer Undergraduate Research Fellowship, Caltech 2022

Steve Tisch Academic Excellence Scholarship, UCLA 2021 – 2024

## TALKS

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Stellar Thinkshop, Carnegie Observatories, Pasadena, CA 2026

Astro Coffee, IAS, Princeton, NJ 2026

Thursday Lunch Talk, Princeton, Princeton, NJ 2026

Stellar-Mass Black Holes Workshop, Talk, KITP 2025

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|------------------------------------------------------------------|------|
| The Lifecycle of Stellar Black Holes Conference, Talk, KITP      | 2025 |
| Carnegie ArXiv Tea, Pasadena, CA                                 | 2025 |
| Keck Science Meeting, Poster, UCLA                               | 2025 |
| Binary Stars in the Space Era, Keele, United Kingdom             | 2025 |
| Vasto Accretion Meeting, Vasto, Italy                            | 2025 |
| AAS Division on Dynamical Astronomy Meeting 56, Atlanta, GA      | 2025 |
| Undergraduate Research Week, UCLA                                | 2024 |
| Exoplanet Monthly Meeting, Northwestern, Evanston, IL            | 2024 |
| APS Conference for Undergraduate Women in Physics, San Diego, CA | 2024 |
| APS Far West Section, UCSD (Helen Quinn Award, 1st)              | 2023 |
| Division 3262 Seminar, NASA/JPL-Caltech                          | 2023 |
| Undergraduate Research Week, UCLA                                | 2023 |
| SURF Final Presentation, Caltech                                 | 2022 |
| HEAG Presentation Week, Caltech                                  | 2022 |

## AWARDED TELESCOPE TIME

|                              |                               |
|------------------------------|-------------------------------|
| JWST                         | 24.8 hours (PI)               |
| Palomar Hale 200 inch, NGPS  | 8 nights (PI) 5 nights (Co-I) |
| Palomar Hale 200 inch, PHARO | 9 nights (PI)                 |
| Keck HIRESb                  | 9 nights (Co-I)               |
| Keck KCWI                    | 1 night (Co-I)                |

## OBSERVING EXPERIENCE

|                                          |       |
|------------------------------------------|-------|
| Palomar Hale Telescope, PHARO – 8 nights | 2024- |
| Keck, HIRES – 2 nights                   | 2025- |

## MENTORING

|                                |       |
|--------------------------------|-------|
| Joe Smith (Caltech Undergrad)  | 2025- |
| Noah East (UCLA Undergrad)     | 2025- |
| Omri Ratzkoff (UCLA Undergrad) | 2025- |

## JOURNAL REFEREE

|                                                  |        |
|--------------------------------------------------|--------|
| ApJ, ApJL, MNRAS, A&A, A&AL ( <i>14 papers</i> ) | 2023 - |
|--------------------------------------------------|--------|

## COLLABORATIONS

|                          |        |
|--------------------------|--------|
| Sloan Digital Sky Survey | 2026 - |
| LISA Consortium          | 2025 - |
| Euclid Consortium        | 2025 - |

## TEACHING & OUTREACH

|                                                                                                                           |                                       |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Graduate Teaching Assistant (Caltech)</b><br>Ay 101: Physics of Stellar Interiors and Atmospheres                      | 2025                                  |
| <b>Graduate Peer Mentor (Caltech)</b>                                                                                     | 2025 -                                |
| <b>LAUSD Guest Speaker</b><br>James Monroe High School<br>Shenendoah Early Education Center                               | 2023 -                                |
| <b>Stargazing Lectures (Caltech)</b><br>Q&A Panelist<br>Telescope Operator                                                | 2023 -                                |
| <b>Diversity, Equity, and Inclusion (DEI) Committee</b><br>Undergraduate Representative, UCLA Physics and Astronomy Dept. | 2023 - 2024<br><i>Los Angeles, CA</i> |
| <b>Undergraduate Learning Assistant</b><br>Physics 1C, Electrodynamics, Optics, and Special Relativity, UCLA              | 2022<br><i>Los Angeles, CA</i>        |

## PUBLICATIONS (19 TOTAL; 12 FIRST-AUTHOR)

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19. Ding, K., Gennaro, M., Avila, R. J., Ricotti, M., Beaton, R. L., Boyer, M. L., Brown, T. M., Calamida, A., Cassisi, S., Chandra, V., Cohen, R. E., Correnti, M., Crnojević, D., El-Badry, K., Geha, M., Guhathakurta, P., Kallivayalil, N., Kirby, E. N., McQuinn, K. B. W., Savino, A., **Shariat, C.**, Simon, J. D., and Weisz, D. R., “Probing the IMF in the Early Universe – Direct measurements in the Boötes I UFD with JWST/NIRCam”, *The Astrophysical Journal*, *submitted*, arXiv:2605.15069, 2026.
  18. **Shariat, C.**, El-Badry, K., MacLeod, M., and Leiner, E., “Gaia astrometry disfavors a binary origin for long secondary periods”, arXiv:2604.09767, Publications of the Astronomical Society of the Pacific, 2026.
  17. **Shariat, C.**, “OverCite: Add citations in LaTeX without leaving the editor”, arXiv:2604.15366, Research Notes of the American Astronomical Society, 2026.
  16. Bhattacharjee, S., El-Badry, K., Fuller, J., **Shariat, C.**, Yamaguchi, N., “Thermally inflated accretors in post-mass transfer binaries: Abell 35 and its class revisited”, arXiv:2603.23756, Publications of the Astronomical Society of the Pacific, 2026.
  15. **Shariat, C.** and El-Badry, K., “A global view of post-interaction white dwarf-main sequence binaries”, Publications of the Astronomical Society of the Pacific, vol. 138, no. 3, Art. no. 034202, 2026.
  14. **Shariat, C.**, Ye, C. S., Naoz, S., and Rose, S., “Fast Radio Bursts from White Dwarf Binary Mergers: Isolated and Triple-Induced Channels”, *The Astrophysical Journal Letters*, vol. 1000, no. 1, Art. no. L17, 2026.
  13. **Shariat, C.**, El-Badry, K., Bhattacharjee, S., “How precisely can we measure the ages of subgiant and giant stars?”, arXiv:2510.08675, *The Open Journal of Astrophysics*, 2026.
  12. Holzknecht, L., Naoz, S., **Shariat, C.**, “Dynamical Pathways to the Misalignment of the VHS 1256-1257 System”, *The Astrophysical Journal*, vol. 998, no. 1, Art. no. 122, 2026.
  11. **Shariat, C.**, El-Badry, K., Gennaro, M., Ding, K., Simon, J.D., Avila, R.J., Calamida, A., Cassisi, S., Correnti, M., Weisz, D.R., Geha, M., Kirby, E.N., Brown, T.M., Ricotti, M., McQuinn, K.B.W., Kallivayalil, N., Gilbert, K., Pacifici, C., Guhathakurta, P., Crnojević, D., Boyer, M.L., Beaton, R.L., Chandra, V., Cohen, R.E., Renzini, A., Savino, A., Tollerud, E.J., “Wide binaries in an ultra-faint dwarf galaxy: discovery, population modeling, and a nail in the coffin of primordial black hole dark matter”, Publications of the Astronomical Society of the Pacific, vol. 137, no. 10, Art. no. 104103, 2025.
  10. Xuan, Z., **Shariat, C.**, Naoz, S., “From Wide Triples to UCXBs: Multimessenger Detection of Dynamically-formed Black Hole-White Dwarf Systems with LISA”, *The Astrophysical Journal*, vol. 995, no. 1, Art. no. 27, 2025.
  9. **Shariat, C.**, El-Badry, K., Naoz, S., “10,000 Resolved Triples from Gaia: Empirical Constraints on Triple Star Populations”, Publications of the Astronomical Society of the Pacific, vol. 137, no. 9, Art. no. 094201, 2025.

8. **Shariat, C.**, El-Badry, K., Naoz, S., Rodriguez, A.C., van Roestel, J., “Cataclysmic Variables in Triples: Formation Models and New Discoveries”, Publications of the Astronomical Society of the Pacific, vol. 137, no. 7, Art. no. 074201, 2025.
7. Matsuno, T., Kemp, A., Tanikawa, A., **Shariat, C.**, El-Badry, K., Dodd, E., Helmi, A., Koch-Hansen, A.J., Yamaguchi, N., Yan, H., “Unevolved Li-rich stars at low metallicity: a possible formation pathway through novae”, Astronomy and Astrophysics, vol. 699, Art. no. A171, EDP, 2025.
6. **Shariat, C.**, Naoz, S., El-Badry, K., Akira Rocha, K., Kalogera, V., Stephan, A.P., Burdge, K., Angelo, I., “Triple Evolution Pathways to Black Hole Low-Mass X-ray Binaries: Insights from V404 Cygni”, The Astrophysical Journal, vol. 983, no. 2, 2025.
5. **Shariat C.**, Naoz, S., El-Badry, K., Rodriguez, A.C., Hansen, B. M. S., Angelo, I., Stephan, A. P., “Once a Triple, Not Always a Triple: The Evolution of Hierarchical Triples that Yield Merged Inner Binaries,” The Astrophysical Journal, vol. 978, no. 1, 2025.
4. Stephan, A. P., Martin, D. V., Naoz, S., Hughes, N. R., and **Shariat, C.**, “Two Novel Hot Jupiter Formation Pathways: How White Dwarf Kicks Shape the Hot Jupiter Population”, The Astrophysical Journal Letters, 2024
3. **Shariat C.**, Hasegawa, Y., Yu, T. Y. M., Hansen, B. M. S., Hu, R., “Predicting the Dominant Formation Mechanism of Multi-Planetary Systems,” The Astrophysical Journal Letters, vol. 964, no. 1, 2024.
2. **Shariat, C.**, Naoz, S., Hansen, B. M. S., Angelo, I., Michaely, E., Stephan, A. P., “Dynamical Evolution of White Dwarfs in Triples in the Era of Gaia,” The Astrophysical Journal Letters, vol. 955, no. 1, 2023.
1. Lazzarini, M., Hinton, K., **Shariat, C.**, Williams, B.F., Garofali, K., Dalcanton, J.J., Durbin, M., Antoniou, V., Binder, B., Eracleous, M., Vulic, N., Yang, J., Wik, D., Gasca, A., Kuauhtzin, Q., “Multiwavelength Characterization of the High-mass X-Ray Binary Population of M33”, The Astrophysical Journal, vol. 952, no. 2, 2023.